

Quantitative Investing

PROBLEM SET 2

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1 QUESTION 1: MEAN-VARIANCE PORTFOLIOS — 10 INDUSTRIES

Data and conventions. The data are monthly value-weighted returns on ten U.S. industry portfolios from July 1926 to June 2026, giving $T = 1200$ observations on $N = 10$ industries, with no missing values. All figures are in percent per month unless stated otherwise: a mean of 1.0584 means 1.0584% per month. The risk-free rate is the sample mean of the monthly risk-free column, $r_f = 0.2701\%$ per month. Short sales are unrestricted, so writing μ for the vector of sample means, Σ for the sample covariance matrix and $\mathbf{1}$ for a vector of ones,

$$w_{\text{MVP}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}, \quad w_{\text{TAN}} = \frac{\Sigma^{-1}(\mu - r_f\mathbf{1})}{\mathbf{1}'\Sigma^{-1}(\mu - r_f\mathbf{1})}, \quad (1)$$

with $\mu_p = w'\mu$, $\sigma_p = \sqrt{w'\Sigma w}$ and $S_p = (\mu_p - r_f)/\sigma_p$.

1.1 Question 1(a): MVP, Tangency Portfolio and the Efficient Frontier

Table 1: Industry sample moments and the resulting portfolio weights from equation (1). Weights sum to one in each column. The last three rows report the mean, standard deviation, Sharpe ratio, gross leverage and cross-sectional weight dispersion of each portfolio.

Industry	Monthly (%)		Weight	
	Mean	Std. Dev.	MVP	Tangency
NoDur	0.9402	4.5595	0.7188	0.6645
Durbl	1.1822	8.2136	-0.0142	0.1340
Manuf	1.0385	6.2008	-0.1905	-0.2242
Enrgy	1.0292	6.4095	0.1527	0.2808
HiTec	1.1699	7.1424	-0.0621	0.1865
Telcm	0.8124	4.6606	0.4889	0.1525
Shops	1.0342	5.7802	-0.0137	0.0725
Hlth	1.0680	5.5005	0.1174	0.3530
Utils	0.8874	5.4351	0.1293	0.0721
Other	0.9237	6.3555	-0.3267	-0.6916
<i>Portfolio mean (%/mo)</i>			0.8672	1.0584
<i>Portfolio std. dev. (%/mo)</i>			3.8346	4.4063
<i>Sharpe ratio (monthly)</i>			0.1557	0.1789
<i>Gross leverage $\sum_i w_i$</i>			2.21	2.83
<i>Cross-sectional SD of weights</i>			0.3085	0.3590

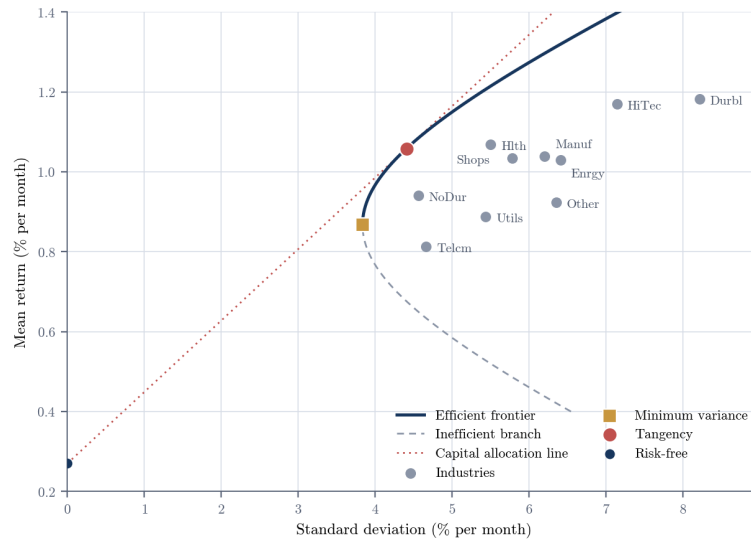


Figure 1: Efficient frontier of the ten industries in mean–standard deviation space. Red dots are the individual industries, the orange square is the minimum variance portfolio, the green star is the tangency portfolio, and the dotted line is the capital allocation line from the risk-free rate. The dashed grey curve is the inefficient lower branch.

Discussion of the weights.

- **NoDur is the largest long position in both portfolios.** It has the lowest standard deviation of the ten industries and, once the other nine are controlled for, the lowest residual volatility, which makes it the most effective hedging instrument in the set. Both objectives want it.
- **The two portfolios disagree most on Telcm.** Telcm has the second-lowest standard deviation and the lowest mean return in the sample. Minimizing variance alone makes it valuable, and once expected return enters, its low volatility no longer pays for the weakest risk premium of the ten.
- **The tangency portfolio is the more aggressive of the two, despite holding more industries long.** It carries higher gross leverage and more dispersed weights. The difference is where the extremity sits: the MVP concentrates on the long side and spreads small shorts across five industries, three of them negligible, while the tangency portfolio funds eight longs out of a single large short in Other, the lowest-Sharpe industry. The extra leverage is what it costs to bet on mean differences that are small next to an average pairwise correlation of 0.696.

Shape of the frontier. We minimize portfolio variance, which is quadratic in the weights, subject to two linear constraints. The optimal weights then come out as an affine function of the target mean, so every frontier portfolio is a fixed combination of two underlying portfolios. This is the two-fund separation result. Substituting that affine function back into the quadratic objective gives a quadratic again, so variance is quadratic in the target mean. The minimum-variance set is therefore a parabola in mean–variance space, and a hyperbola in the mean–standard deviation space of Figure 1. Two things follow.

- **The curvature is diversification.** If the industries were perfectly correlated, combining them would reduce no risk and the attainable set would be a straight line. Correlations below

one bend it leftward, and how far it bends measures how much diversification is available. Here the bend is modest: the MVP is only about a sixth less volatile than the least volatile industry, where part (c) shows it would be less than half as volatile if the industries were uncorrelated.

- **Dominance makes only the upper branch efficient.** Because variance is quadratic in the target mean, every variance above the minimum is attained by two portfolios, one on each side of the minimum variance portfolio. They carry identical risk but the lower one returns less, so it is dominated. Since investors prefer a higher expected return at the same risk, no one would hold it: the lower branch is feasible but never optimal, which is why it is dashed in Figure 1.

1.2 Question 1(b): Reliability of the Mean Return Estimates

The standard error of each sample mean is $\hat{\sigma}_i/\sqrt{T}$ with $T = 1200$.

Table 2: Reliability of the estimated mean returns, annualized. The t -statistic tests the null that the true mean return is zero.

Industry	Mean (%/yr)	Std. Error (%/yr)	t -statistic
NoDur	11.28	1.58	7.14
Durbl	14.19	2.85	4.99
Manuf	12.46	2.15	5.80
Enrgy	12.35	2.22	5.56
HiTec	14.04	2.47	5.67
Telcm	9.75	1.61	6.04
Shops	12.41	2.00	6.20
Hlth	12.82	1.91	6.73
Utils	10.65	1.88	5.66
Other	11.08	2.20	5.03

We then raise every estimated mean by one of its own standard errors, $\tilde{\mu}_i = \hat{\mu}_i + \hat{\sigma}_i/\sqrt{T}$, and recompute the tangency portfolio. The MVP is unchanged by construction because it does not use μ .

Table 3: Tangency portfolio weights and statistics before and after raising every estimated mean by one standard error.

Industry	Baseline	Means +1 SE	Change
NoDur	0.6645	0.5686	−0.0960
Durbl	0.1340	0.1526	0.0187
Manuf	−0.2242	−0.2611	−0.0370
Enrgy	0.2808	0.2864	0.0056
HiTec	0.1865	0.1781	−0.0084
Telcm	0.1525	0.1709	0.0184
Shops	0.0725	0.0671	−0.0054
Hlth	0.3530	0.3400	−0.0129
Utils	0.0721	0.0965	0.0244
Other	−0.6916	−0.5991	0.0926
<i>Portfolio mean (%/mo)</i>	1.0584	1.1980	0.1395
<i>Portfolio std. dev. (%/mo)</i>	4.4063	4.3771	−0.0292
<i>Sharpe ratio (monthly)</i>	0.1789	0.2120	0.0331

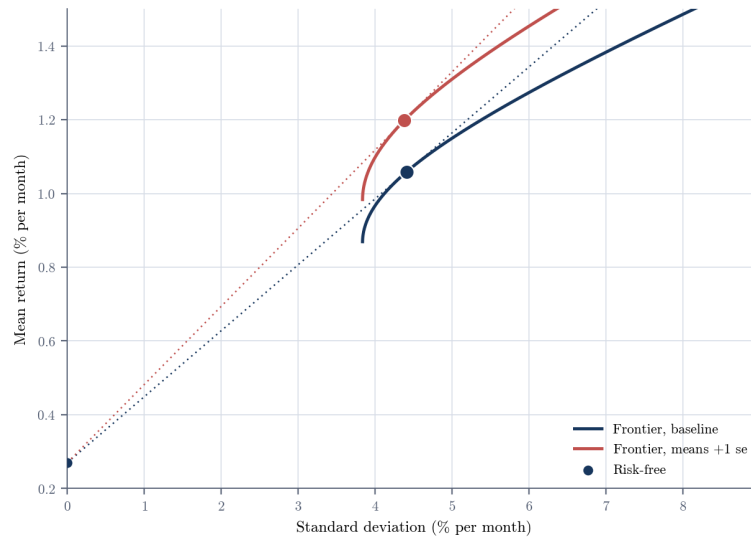


Figure 2: Efficient frontiers and capital allocation lines under the baseline means (blue) and after a one-standard-error increase in every mean (red). Stars mark the tangency portfolio in each case.

Discussion.

- **The means are not estimated reliably enough to rank the industries.** Every industry mean differs from zero with a comfortable t -statistic, but optimization needs the ranking rather than the level. Of the 45 pairwise differences only one is significant at the 5% level, and 45 tests at that level would produce about two false positives by chance, so the cross-section is indistinguishable from ten industries with identical means. Splitting the century at 1976 adds nuance rather than comfort: the extremes hold, with Durbl and HiTec leading both halves and Telcm trailing both, while the middle reshuffles.
- **The tangency weights move only moderately.** Raising every mean by one standard error tilts the composition without turning it over; the two weight vectors keep a cosine similarity above 0.99.
- **The frontier moves much more than the weights do.** The tangency Sharpe ratio rises by almost a fifth, and the capital allocation line shifts with it (Figure 2). So a perturbation that no test could detect relocates the whole opportunity set. We raised every mean at once, so a good part of this is the frontier shifting up as a whole. That is still the point: a century of data does not pin down even the level of the achievable Sharpe ratio.

1.3 Question 1(c): Reliability of the Covariance Matrix

We recompute the frontier with the covariances set to zero (keeping the estimated variances on the diagonal) and then with Σ replaced by the identity matrix. Table 4 reports each portfolio twice: once with risk measured by the covariance matrix used to build it, and once with the same weights evaluated under the true sample Σ . The second measurement is needed because the assumed risk units are not comparable across the three cases, and under the identity matrix they are arbitrary.

Table 4: Portfolio statistics under the three covariance assumptions. The mean vector μ is the same in every case. “Assumed” measures risk with the matrix used to construct the weights; “true” measures the same weights under the full sample covariance matrix. Means and standard deviations are monthly percentages.

Case	Mean	Assumed Σ		True Σ		Loss
		Std. Dev.	Sharpe	Std. Dev.	Sharpe	
Full sample — MVP	0.8672	3.8346	0.1557	3.8346	0.1557	0.0%
Full sample — Tangency	1.0584	4.4063	0.1789	4.4063	0.1789	0.0%
Diagonal — MVP	0.9794	1.8242	0.3888	4.8467	0.1463	−6.0%
Diagonal — Tangency	0.9960	1.8455	0.3933	4.9516	0.1466	−18.1%
Identity — MVP	1.0086	0.3162	2.3351	5.1422	0.1436	−7.8%
Identity — Tangency	1.0257	0.3199	2.3621	5.2675	0.1434	−19.8%

Table 5: Decomposition of portfolio variance into the contribution of the ten variance terms and the forty-five covariance terms, using the full sample covariance matrix. Monthly, in squared percent.

Portfolio	Total variance	From variances	From covariances	Covariance share
Equal weighted	26.442	3.741	22.701	85.9%
MVP	14.704	23.725	−9.021	−61.3%
Tangency	19.415	41.264	−21.849	−112.5%

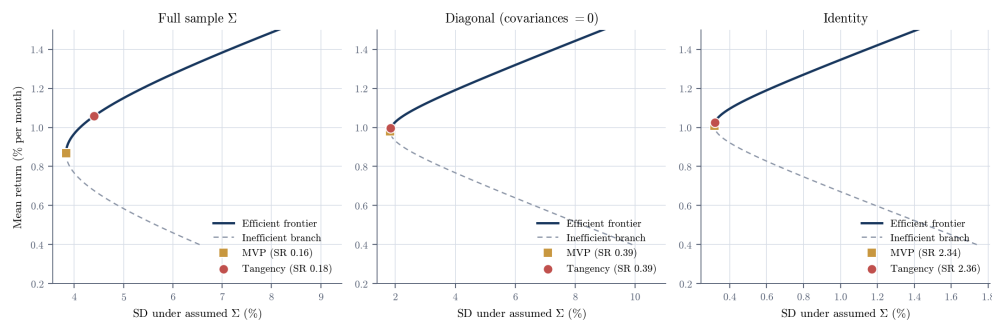


Figure 3: Efficient frontiers under the full sample, diagonal and identity covariance matrices, each drawn in its own risk units. The horizontal axes are therefore not comparable across panels.

Discussion.

- **The covariance matrix is estimated precisely, though it is not stable.** A mean carries roughly eight times the relative error of a volatility, because a mean’s precision depends on how far returns bounce relative to how far they drift, while a volatility’s depends only on the sample size. Daily data would sharpen Σ further and would not help μ at all. Stability is a separate matter: splitting the century at 1976, average pairwise correlation falls from 0.779 to 0.581, so Σ is measured precisely and still moves. The two experiments below set 45 covariances to zero, which is far bigger than any estimation error, so they test what happens when we assume something wrong rather than when we measure something badly.
- **The frontier changes far more than in part (b).** Measured on its own assumed risk, the tangency Sharpe ratio more than doubles under the diagonal matrix and rises more than tenfold under the identity, against a move of under a fifth in part (b). Applying those same weights and measuring risk with the real Σ reverses the ordering entirely. Dropping the covariances makes the portfolio look much better and leaves it slightly worse.

- **The covariance terms matter far more than the variance terms.** For an equal-weighted portfolio, the great majority of total variance comes from the 45 covariance terms rather than the 10 variances. Two results confirm it.

The first is the shape of the two misspecified frontiers, which land within a few per cent of each other and need roughly twice the relative risk of the real frontier to reach the same distance above the MVP. Removing the correlations changes everything, and flattening the variances afterwards changes almost nothing. With real correlations the ten industries act like 2.26 independent bets, and with correlations removed they act like 10.

The second is that the damage falls unevenly. Misspecifying Σ costs the tangency portfolio about three times what it costs the MVP, and wipes out its entire advantage over the MVP. Minimizing risk works roughly well with variances alone, so the MVP survives, while the tangency portfolio's whole edge comes from the covariances.

1.4 Question 1(d): Jorion Simulation under the Normal Distribution

Each simulation draws $T = 1200$ monthly return vectors from a multivariate normal distribution with the historical μ and Σ . The MVP and tangency weights are computed from the *simulated* sample and then applied to the *actual* industry returns, so the recorded mean and standard deviation are $w'\mu$ and $\sqrt{w'\Sigma w}$ at the historical moments. This is the essential feature of the Jorion (1986) design: the weights carry estimation error but are judged against the true distribution. The experiment is repeated 1,000 times.

Table 6: Estimation error across 1,000 simulations, for both Question 1(d) and Question 1(e). Portfolio means and standard deviations are evaluated on the actual industry returns, in percent per month. “SD of weights” is the cross-simulation standard deviation of a portfolio weight, averaged over the ten industries.

Experiment	Portfolio mean		Portfolio std. dev.		SD of weights
	Average	Std. dev.	Average	Std. dev.	
Normal — MVP	0.8671	0.0096	3.8489	0.0066	0.0381
Normal — Tangency	1.0650	0.0793	4.9938	0.6388	0.2716
Bootstrap — MVP	0.8693	0.0176	3.8627	0.0142	0.0523
Bootstrap — Tangency	1.0692	0.0802	5.0046	0.6377	0.2703

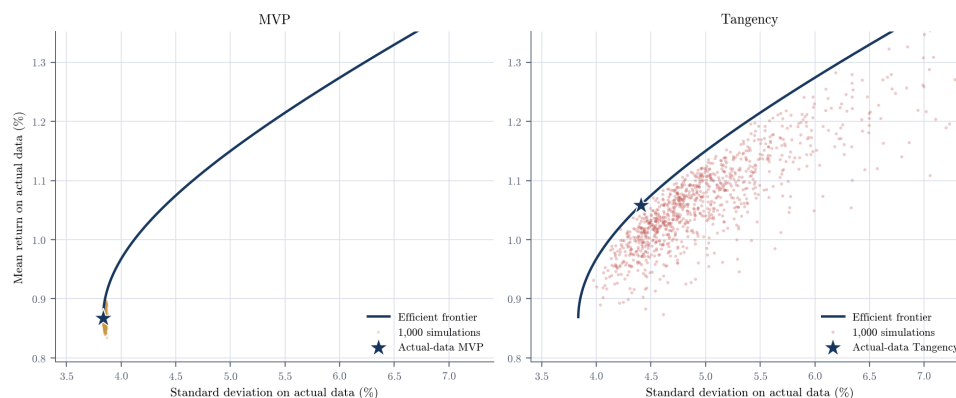


Figure 4: Question 1(d). Each point is one of 1,000 simulations: weights estimated from a multivariate normal sample of $T = 1200$ months, then applied to the actual industry returns. The black star is the portfolio built from the actual data. Both panels share the same axes.

Discussion.

- **The MVP is estimated with far less error than the tangency portfolio.** All three measures in Table 6 agree. The MVP is tighter on the spread of the weights, the spread of the realized mean and the spread of the realized standard deviation, by roughly a factor of eight on the first two and two orders of magnitude on the third. Figure 4 shows the same thing: the MVP simulations sit on top of the actual-data portfolio, while the tangency simulations scatter well to the right of the frontier.
- **The tangency portfolio needs the noisy input, and the MVP sits at the minimum of the risk surface.** The MVP uses only the covariance matrix, while the tangency portfolio also needs the mean vector, which part (b) showed is the noisy one. The MVP also minimizes variance, so the slope of risk with respect to the weights is zero there and a weight error costs it only a second-order amount. That is why the gap widens from the weights to the realized volatility.
- **The tangency portfolio still earns a higher Sharpe ratio, but estimation error removes most of its advantage.** It wins in about three quarters of the simulations. Its advantage falls from roughly fifteen per cent on paper to about three per cent once estimation error is taken into account.

AI Integration — Simulation code generation.

- **Prompt.** We gave ChatGPT the spreadsheet and the following prompt.

You're an expert finance analyst. You need to write a Python code file that includes steps as followed:

1. Draw T observations from a multivariate normal distribution with estimated mean vector and covariance matrix;
2. Computes tangency and MVP weights from the simulated data
3. Applies those weights to the actual, not simulated, industry return matrix.
4. Record the resulting portfolio mean and standard deviation. Repeat 1000 times.

The output would be a python file, and resulted mean and st report with a good visualization and formatting. At the end, tell me: which portfolio has smaller estimation error, and what is the economic explanation?

- **What we had to fix or adjust.** The data output is sound. But because we didn't specify a format for the chart, its figure does not match what the question asks for. It marks the actual-data portfolios, but draws one combined scatter instead of the two separate mean-standard deviation plots the question specifies, and it shows no efficient frontier. Extreme draws also stretch the axes, most visibly on the volatility-error histogram, where the scale runs past five percentage points while almost every draw sits below two. Our own code for part (d), producing Figure 4, uses two shared-axis panels, marks the actual-data portfolios, adds the efficient frontier, and clips the axis limits by percentile so one outlier draw cannot rescale the plot.

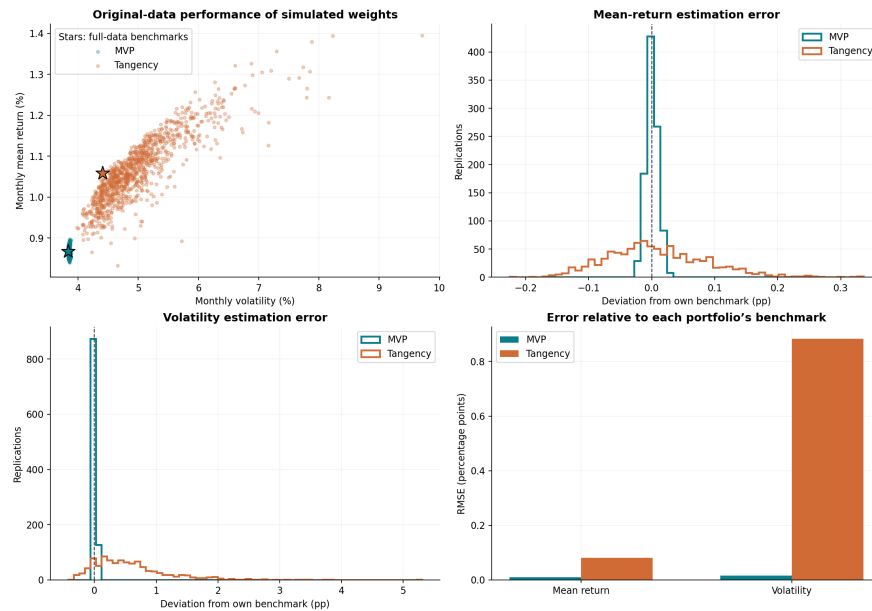


Figure 5: The chart produced by ChatGPT’s script, reproduced from its own HTML report. Stars mark the full-data benchmark portfolios.

A small data analysis difference: it re-factorizes the covariance matrix on every draw rather than reusing a single Cholesky factor, which is correct but slower.

- **Did the LLM correctly apply the simulated weights to the actual returns?** Yes. Jorion’s logic implies that no simulated portfolio should ever beat the true optimum once judged against the original mean and covariance, and none did for either portfolio in this test. If weights were evaluated on the simulated sample, it would have failed this test constantly; the LLM got the key distinction right here.
- **Does the LLM correctly predict which portfolio has smaller estimation error, and does its economic explanation hold up?**
 - **Prediction.** Correct. ChatGPT’s own generated report reproduces the same asymmetry we found (Table 7): the MVP has the smaller estimation error in both mean and volatility, matching our own Table 6 and Figure 4.

Table 7: Estimation-error metrics as reported in ChatGPT’s own generated output, `portfolio_simulation_report.html`.

Metric	MVP	Tangency
Mean-return RMSE (pp)	0.0095	0.0817
Volatility RMSE (pp)	0.0160	0.8827
Weight RMSE (pp of allocation)	4.0318	28.2099

- **Explanation.** Also correct. It attributes the gap to the MVP depending only on the covariance matrix while the tangency portfolio additionally depends on the noisy mean vector. It also separately notes that MVP volatility sits at a variance minimum, so weight errors cost only a second-order amount.

1.5 Question 1(e): Empirical Row Bootstrap Simulation

We repeat the experiment under the empirical distribution. Each simulation draws $T = 1200$ month indices uniformly with replacement from $\{1, \dots, T\}$ and takes the corresponding rows of the actual return matrix, so all ten industry returns for a sampled month stay together. Sampling whole rows preserves the contemporaneous cross-sectional dependence and the non-normal shape of the marginal distributions, while destroying serial dependence such as volatility clustering. As before, weights are estimated on the bootstrap sample and applied to the actual returns, 1,000 times. Results are in Table 6 alongside the normal simulation.

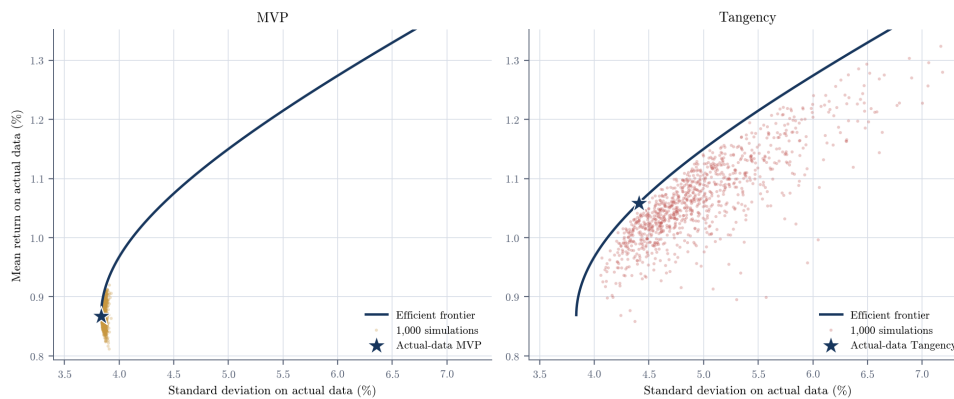


Figure 6: Question 1(e). Weights estimated on empirical row-bootstrap samples and applied to the actual industry returns, 1,000 simulations. Axes are identical to Figure 4 so the two experiments can be compared directly.

Discussion.

- **The bootstrap raises estimation error for the MVP and leaves the tangency portfolio unchanged.** Table 6 shows the MVP spreading roughly twice as wide under the bootstrap on both the realized mean and the realized standard deviation, while the tangency figures are effectively identical under the two samplers. We repeated the comparison across twenty independent seed pairs and the split holds every time.
- **Fat tails explain the split, because they affect the two inputs differently.** Monthly industry returns are strongly non-normal: Jarque-Bera rejects normality for all ten industries and average excess kurtosis is above seven. The sampling variance of a mean depends only on the second moment, so fat tails leave the mean vector exactly as precise as it was. The sampling variance of a variance carries a term in excess kurtosis, which here roughly doubles its standard error. The bootstrap therefore makes the covariance matrix about twice as noisy and does nothing at all to the mean vector.
The MVP uses only the covariance matrix, so it absorbs that inflation in full, and its extra dispersion comes out close to what the kurtosis predicts. For the tangency portfolio we split the error into a part driven by the mean vector and a part driven by the covariance matrix, and almost all of it comes from the mean. The bootstrap inflates the smaller part, which barely moves the total.
- **The conclusion from part (d) survives.** The tangency cloud remains far wider than the MVP cloud under the bootstrap, so switching to the empirical distribution narrows the gap without coming close to closing it. One caveat: the row bootstrap samples months independently, so it reproduces fat tails and contemporaneous correlation while destroying volatility clustering. It captures one way the normal assumption fails and misses another.

2 QUESTION 2: THREE-ASSET HAND CALCULATION

Setup. Three assets A, B, C have mean returns $\mu_A = 17\%$, $\mu_B = 13\%$, $\mu_C = 9\%$ and standard deviations $\sigma_A = 20\%$, $\sigma_B = 40\%$, $\sigma_C = 15\%$. The pairwise correlations are $\rho_{AB} = 0.50$, $\rho_{BC} = 0.30$, $\rho_{AC} = 0$, and the risk-free rate is $r_f = 5\%$. All quantities below are decimals (e.g. $17\% = 0.17$). Using $\text{Cov}(i, j) = \rho_{ij}\sigma_i\sigma_j$,

$$\Sigma = \begin{pmatrix} 0.0400 & 0.0400 & 0 \\ 0.0400 & 0.1600 & 0.0180 \\ 0 & 0.0180 & 0.0225 \end{pmatrix}, \quad \Sigma^{-1} \approx \begin{pmatrix} 34.4697 & -9.4697 & 7.5758 \\ -9.4697 & 9.4697 & -7.5758 \\ 7.5758 & -7.5758 & 50.5051 \end{pmatrix}.$$

2.1 Question 2(a): MVP and Tangency Portfolio

Using $w_{\text{MVP}} = \Sigma^{-1}\mathbf{1}/(\mathbf{1}'\Sigma^{-1}\mathbf{1})$ with $\Sigma^{-1}\mathbf{1} = (32.5758, -7.5758, 50.5051)$ and $\mathbf{1}'\Sigma^{-1}\mathbf{1} = 75.5051$,

$$w_{\text{MVP}} = (0.4314, -0.1003, 0.6689), \quad \mu_{\text{MVP}} = 12.05\%, \quad \sigma_{\text{MVP}} = 11.51\%.$$

Using $w_{\text{TAN}} = \Sigma^{-1}(\mu - r_f\mathbf{1})/[\mathbf{1}'\Sigma^{-1}(\mu - r_f\mathbf{1})]$ with $\mu - r_f\mathbf{1} = (0.12, 0.08, 0.04)$, $\Sigma^{-1}(\mu - r_f\mathbf{1}) = (3.6818, -0.6818, 2.3232)$ and $\mathbf{1}'\Sigma^{-1}(\mu - r_f\mathbf{1}) = 5.3232$,

$$w_{\text{TAN}} = (0.6917, -0.1281, 0.4364), \quad \mu_{\text{TAN}} = 14.02\%, \quad \sigma_{\text{TAN}} = 13.02\%.$$

Table 8: MVP and tangency portfolio weights and moments for the three-asset problem.

	MVP	Tangency
w_A	43.14%	69.17%
w_B	-10.03%	-12.81%
w_C	66.89%	43.64%
<i>Mean</i>	12.05%	14.02%
<i>Std. dev.</i>	11.51%	13.02%

2.2 Question 2(b): Equation of the Efficient Frontier

With $A = \mathbf{1}'\Sigma^{-1}\mathbf{1} = 75.5051$, $B = \mathbf{1}'\Sigma^{-1}\mu = 9.0985$, $C = \mu'\Sigma^{-1}\mu = 1.2013$ and $D = AC - B^2 = 7.9209$, the efficient frontier is

$$\sigma_p^2 = \frac{1}{A} + \frac{A}{D} \left(\mu_p - \frac{B}{A} \right)^2 = 0.013244 + 9.5324 (\mu_p - 0.1205)^2, \quad (2)$$

where $B/A = 0.1205$ is exactly the mean of the MVP found in part (a). The efficient portion is the upper branch, $\mu_p \geq 12.05\%$.

2.3 Question 2(c): Target Return and Target Risk, Risky Assets Only

Minimum variance for $\mu_p = 13\%$. Writing any frontier portfolio as $w = \alpha w_{\text{MVP}} + (1 - \alpha) w_{\text{TAN}}$ and solving $0.13 = \alpha(0.1205) + (1 - \alpha)(0.1402)$ gives $\alpha = 0.5180$, so

$$w = (0.5569, -0.1137, 0.5569), \quad \sigma_p = \sqrt{0.013244 + 9.5324(0.13 - 0.1205)^2} = 11.88\%.$$

$$\text{Sharpe} = \frac{0.13 - 0.05}{0.1188} = 0.674.$$

Maximum return for $\sigma_p = 15\%$. Solving Equation (2) at $\sigma_p = 0.15$ on the efficient branch gives $\mu_p = 15.17\%$, and $0.1517 = \alpha(0.1205) + (1 - \alpha)(0.1402)$ gives $\alpha = -0.5812$, so

$$w = (0.8429, -0.1442, 0.3013), \quad \text{Sharpe} = \frac{0.1517 - 0.05}{0.15} = 0.678.$$

2.4 Question 2(d): Adding the Risk-free Asset

With the risk-free asset available, every efficient portfolio combines r_f with the tangency portfolio along the capital allocation line

$$\mu_p = r_f + \frac{\mu_{\text{TAN}} - r_f}{\sigma_{\text{TAN}}} \sigma_p = 0.05 + 0.693 \sigma_p.$$

Minimum variance for $\mu_p = 13\%$. Solving $0.13 = 0.05 + y(0.1402 - 0.05)$ gives $y = 0.8868$ invested in the tangency portfolio and $1 - y = 11.32\%$ in the risk-free asset, so $w = (61.34\%, -11.36\%, 38.70\%)$ in A, B, C , $\sigma_p = 0.8868(0.1302) = 11.54\%$, and Sharpe = 0.693.

Maximum return for $\sigma_p = 15\%$. Solving $0.15 = y(0.1302)$ gives $y = 1.1523$: the investor borrows 15.23% at the risk-free rate to lever the tangency portfolio beyond full investment, giving $w = (79.70\%, -14.76\%, 50.29\%)$, $\mu_p = 15.39\%$, and Sharpe = 0.693.

Table 9: Risky-assets-only portfolios from part (c) compared with the risk-free-augmented portfolios of part (d), at matched target mean and matched target standard deviation.

	Target $\mu_p = 13\%$		Target $\sigma_p = 15\%$	
	(c) risky only	(d) with r_f	(c) risky only	(d) with r_f
Mean	13.00%	13.00%	15.17%	15.39%
Std. dev.	11.88%	11.54%	15.00%	15.00%
Sharpe ratio	0.674	0.693	0.678	0.693

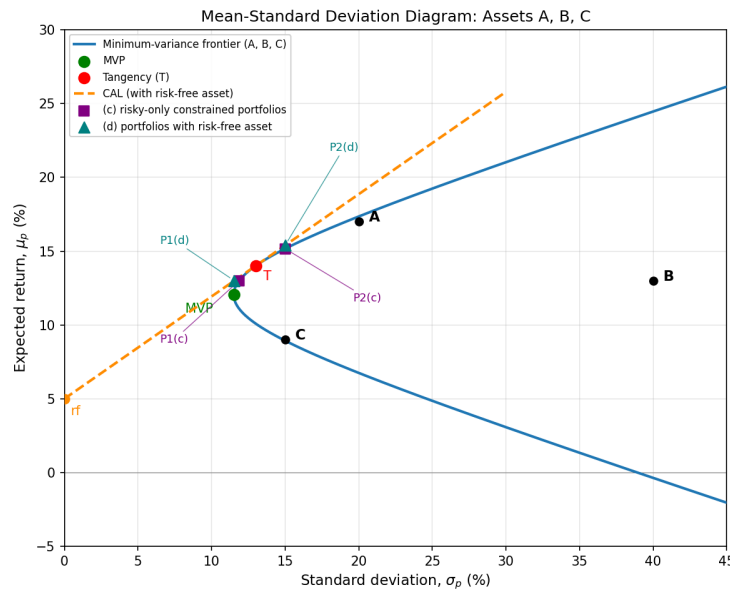


Figure 7: Mean–standard deviation diagram for the three-asset problem: the risky-asset efficient frontier from part (b), the tangency portfolio, and the capital allocation line from r_f through the tangency portfolio that becomes available once the risk-free asset is added in part (d).

Discussion.

- **Adding the risk-free asset weakly dominates the risky-only frontier everywhere except at the tangency portfolio.** For the same target mean of 13%, allowing the risk-free asset lowers risk from 11.88% to 11.54%; for the same target risk of 15%, it raises the attainable mean from 15.17% to 15.39%. The capital allocation line is tangent to the risky-asset frontier at the tangency portfolio and lies above the frontier everywhere else, so introducing r_f can only help (Figure 7).
- **Both risk-free-augmented portfolios share the tangency portfolio's Sharpe ratio of 0.693.** Every portfolio on the capital allocation line is a combination of the risk-free asset and the tangency portfolio, so the ratio of its risk premium to its risk is pinned at the tangency portfolio's Sharpe ratio regardless of how much is borrowed or lent.
- **The two target-risk portfolios differ in kind, not just degree.** Meeting $\mu_p = 13\%$ needs $y = 0.887 < 1$, so part of the portfolio is lent out at the risk-free rate; meeting $\sigma_p = 15\%$ needs $y = 1.152 > 1$, so the investor borrows at the risk-free rate to lever the tangency portfolio beyond one hundred percent risky exposure.

3 QUESTION 3: ROBUST PORTFOLIO CONSTRUCTION

3.1 Step 1: Evaluating Three Robust Methods

Ledoit–Wolf covariance shrinkage. Ledoit–Wolf shrinkage combines the sample covariance matrix with a simpler target, usually a scaled identity matrix. Shrinking the covariance estimate toward this target reduces the effect of sampling noise and can produce more stable portfolio weights. *Evaluation:* the LLM's explanation is correct. One nuance it misses is that shrinkage affects the entire covariance matrix, including its eigenvalues, rather than only the off-diagonal covariance estimates. The standard Ledoit–Wolf method also estimates the shrinkage intensity from the data, so it generally should not be chosen manually.

Minimum-variance portfolio with Ledoit–Wolf covariance. This approach constructs the minimum-variance portfolio using a Ledoit–Wolf covariance matrix in place of the ordinary sample covariance matrix. It requires no expected-return estimates, so it avoids one of the largest sources of error in the classical tangency portfolio. *Evaluation:* the LLM's explanation is correct, but the method does not remove estimation error completely, since the covariances must still be estimated and may vary over time. It can also produce large or negative weights if short selling is allowed and no position or turnover constraints are imposed.

Black–Litterman. Black–Litterman begins with expected returns implied by the market portfolio and combines them with the investor's own views in a Bayesian framework. Compared with using historical average returns directly, it usually produces more stable expected-return estimates and less extreme portfolio weights. *Evaluation:* the LLM's explanation is correct but incomplete: it does not flag that Black–Litterman is not parameter-free. It requires market-cap or benchmark weights, a risk-aversion assumption, investor views, and confidence levels for those views, and the result depends on these choices. Because our dataset has no market-cap weights for the ten industries, we treat Black–Litterman as a research-only option here rather than something we could implement without further assumptions.

3.2 Step 2: Chosen Method — Minimum-Variance Portfolio with Ledoit–Wolf Covariance

Choice of method. We chose the minimum-variance portfolio with Ledoit–Wolf covariance because it addresses both sources of instability identified in Question 1: it improves covariance estimation through shrinkage, and it avoids noisy expected-return estimates entirely. Compared with using Ledoit–Wolf covariance inside a mean-return-based portfolio, the MVP removes the additional instability that estimated expected returns would otherwise contribute. We did not choose Black–Litterman because the dataset does not provide the market-cap weights or investor views its implementation requires.

Prompt. We gave the LLM the spreadsheet and the following prompt.

Using the monthly returns for the ten industries, write concise Python code that estimates a Ledoit–Wolf covariance matrix and constructs a fully invested minimum-variance portfolio. Allow short selling so that the constraints are consistent with the unconstrained classical tangency portfolio. Estimate the shrinkage intensity automatically, report the robust portfolio weights, and compare their dispersion with the classical tangency weights. Explain the parameter and constraint choices and include checks that the optimization was completed correctly.

Parameter and constraint choices. We used the standard `LedoitWolf` estimator, so the shrinkage intensity was estimated automatically rather than chosen by hand. The portfolio was constrained to be fully invested (weights summing to one), while short selling was left unrestricted to stay consistent with the unconstrained tangency portfolio of Question 1(a). We used `SLSQP` for the minimum-variance optimization from an equal-weight starting point, and converted returns from percentages to decimals before estimating either covariance matrix.

```
lw = LedoitWolf().fit(R)
cov_lw = lw.covariance_

result = minimize(
    lambda w: w @ cov_lw @ w,
    np.ones(n) / n,
    method="SLSQP",
    constraints={"type": "eq", "fun": lambda w: w.sum() - 1},
    options={"ftol": 1e-14, "maxiter": 2000},
)
w_lw = result.x
```

Table 10: Classical tangency weights (Question 1(a)) versus the Ledoit–Wolf minimum-variance weights. The Ledoit–Wolf shrinkage intensity was estimated at 0.0139.

Industry	Classical tangency (%)	LW minimum variance (%)	Difference (pp)
NoDur	66.45	64.65	−1.81
Durbl	13.40	−2.41	−15.81
Manuf	−22.42	−16.46	5.96
Enrgy	28.08	14.99	−13.10
HiTec	18.65	−7.26	−25.90
Telcm	15.25	48.38	33.14
Shops	7.25	1.18	−6.07
Hlth	35.30	13.08	−22.22
Utils	7.21	13.91	6.71
Other	−69.16	−30.07	39.10

Table 11: Weight dispersion: classical tangency versus Ledoit–Wolf minimum variance.

	Classical tangency	LW minimum variance
Minimum weight (%)	−69.16	−30.07
Maximum weight (%)	66.45	64.65
Weight std. dev. (pp)	35.90	28.57
Gross exposure $\sum_i w_i $	2.83	2.12

Corrections and validation. No substantive correction to the portfolio methodology was required. We used a tight optimization tolerance for numerical precision and validated the result three ways: the weights sum to one, the Ledoit–Wolf covariance matrix has strictly positive eigenvalues, and the optimized weights match the closed-form minimum-variance solution $\Sigma_{\text{LW}}^{-1}\mathbf{1}/(\mathbf{1}'\Sigma_{\text{LW}}^{-1}\mathbf{1})$ to within 10^{-6} .

Results. The Ledoit–Wolf estimator produced a shrinkage intensity of only 0.0139, so it places nearly all its weight on the sample covariance matrix; with $T = 1200$ observations for ten industries this is reasonable, though it does not mean the sample covariance matrix is free of estimation error. The Ledoit–Wolf MVP is nonetheless less extreme than the classical tangency portfolio (Table 11): the cross-sectional weight standard deviation falls from 35.90 to 28.57 percentage points, gross exposure falls from 2.83 to 2.12, and the most negative position moves from −69.16% to −30.07%. Substantial long and short positions remain because short selling is still unrestricted. Importantly, this difference should not be attributed to covariance shrinkage alone: the tangency portfolio depends on both estimated expected returns and covariances, while the Ledoit–Wolf MVP minimizes variance without using expected returns at all, so the comparison reflects the stability of the whole robust construction rather than the isolated effect of shrinkage.

3.3 Step 3: Empirical Bootstrap Comparison

We reuse the empirical row-bootstrap engine from Question 1(e): each simulation draws $T = 1200$ month indices with replacement, re-estimates both the classical tangency portfolio and the Ledoit–Wolf MVP on the resampled data, and evaluates the resulting weights against the actual industry-return moments, 1,000 times.

Table 12: Bootstrap estimation error, classical tangency versus Ledoit–Wolf minimum variance, 1,000 simulations. “Weight SD” is the cross-simulation standard deviation of a portfolio weight, averaged over the ten industries.

	Classical tangency	LW minimum variance
Average weight SD (pp)	27.03	4.83
SD of realized mean (%/mo)	0.0802	0.0168
SD of realized std. dev. (%/mo)	0.6377	0.0156

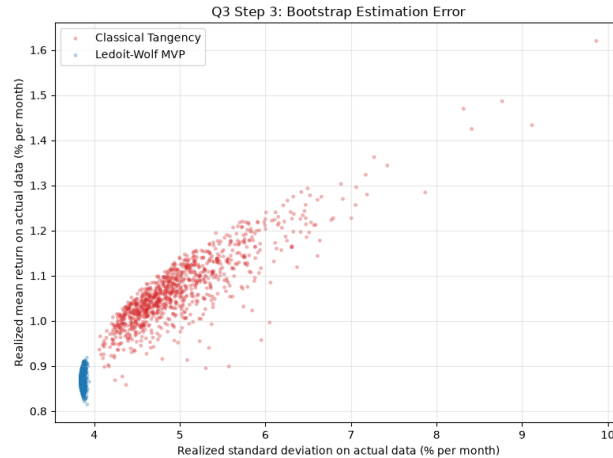


Figure 8: Bootstrap estimation error: each point is one of 1,000 bootstrap resamples, with weights estimated on the resample and evaluated on the actual industry returns. The Ledoit–Wolf MVP cloud (blue) is far tighter than the classical tangency cloud (red).

Discussion.

- **The Ledoit–Wolf MVP is substantially more stable than the classical tangency portfolio across all three measures.** Average cross-simulation weight SD falls from 27.03 to 4.83 percentage points, the SD of realized mean returns falls from 0.0802% to 0.0168% per month, and the SD of realized volatility falls from 0.6377% to 0.0156% per month.
- **The gain has two sources, not one.** The result is consistent with the tangency portfolio depending on noisy expected-return estimates in addition to covariance estimates, while the Ledoit–Wolf MVP avoids expected-return estimation altogether and uses a shrinkage-based covariance estimate. The improvement should not be attributed to Ledoit–Wolf shrinkage alone: switching from a tangency objective to an MVP objective also removes the especially noisy expected-return estimates on its own.

3.4 Step 4: Reflection

The Ledoit–Wolf MVP still relies on historical covariance estimates and leaves out expected returns entirely, so its greater stability may come at the cost of failing to exploit persistent return differences across industries. Shrinkage may also oversimplify the covariance structure if the historical relationships it pulls away from actually contain useful information. We would expect the method to deliver more stable out-of-sample performance when expected returns are hard to estimate, while it may underperform a well-specified tangency strategy when expected-return signals are persistent and reliable. In practice, we would use it as a robust baseline or core allocation, combined with position and turnover limits and regular out-of-sample monitoring, rather than as a universally optimal portfolio.